Item	Conditions	Rating
Traffic Control Plan	The Traffic Control Plan meets the Ministry's minimum requirements for the Project Category identified in the contract or in the Project Category determination process. It includes text descriptions of the location of the work zone, proposed work activities, proposed traffic control measures, and the specific times and dates when work will be undertaken on the roadway. It includes traffic control layouts (schematic diagrams) of the roadway showing the placement and general arrangement of traffic control devices. It includes customized layouts where standard layouts are inadequate. Layouts need not be to scale, but should include dimensions and site-specific characteristics. (Drawings are required for a Traffic Control Plan only if standard layouts are not adequate.) It may include customized drawings of the roadway in the vicinity of the work zone that identify the arrangement of traffic control devices in accordance with the standards identified in Sections 6 through 19 (Traffic Control Layouts) of the Ministry's Traffic Management Manual for Work on Roadways. The drawings include dimensions, and show all painted markings, physical features that may affect traffic operations (signing, guard rails, lamp standards, etc.), road geometry, and lane configurations. The Traffic Control Plan is detailed to the extent appropriate for the complexity of the work or incident. It was prepared by the Prime Contractor, and was shared with all responsible parties before the commencement of the work.	

Item	Conditions	Rating
Incident Management Plan	For large, complex projects, the Incident Management Plan addresses procedures for handling unplanned events and incidents, and includes provisions for incident response. The Incident Management Plan specifically identifies: • types of traffic incidents that may occur in the work zone and planned estimated resumption times • procedures for detecting and verifying incidents that occur within the work zone • procedures for responding to incidents • procedures for restoring traffic flow around an incident site as quickly as possible, including the use of detours • procedures for clearing the incident and restoring pre-incident traffic operations as soon as possible • procedures for identifying estimated resumption times • procedures for immediately informing the Road Authority of the incident occurrence, response measures taken, clearance measures planned, estimated clearance time, and actual incident clearance time • procedures for informing major user groups and the travelling public of anticipated delays and the estimated duration of the unplanned traffic pattern change • procedures for reviewing incidents and proposing modifications to the Traffic Management Plan that will enhance the work site's safety and ease of use • duties and responsibilities of the Traffic Control Manager, Traffic Control Supervisor, and Traffic Control Persons with respect to incident response operations • a contact list for emergency response agencies	

Item	Conditions	Rating
Public Information Plan	The Public Information Plan identifies: • major stakeholders and road users that may be affected by the work • types of communication processes that will be used to notify stakeholders and road users of the work • actions/procedures for informing the Road Authority, travelling public, and project stakeholders of current traffic operations and planned changes to traffic operations • actions/procedures for informing major user groups and the travelling public of anticipated delays • actions/procedures for informing major user groups of the estimated duration of unplanned traffic pattern changes	
Implementation Plan	The Implementation Plan identifies: • project phases and milestones for periods when traffic operations may change • specific changes required to traffic operations during various phases of the work • responsibilities and procedures for ensuring that the sub-plans in the Traffic Management Plan are implemented • qualifications, responsibilities, and duties of the management and supervisory personnel who are responsible for implementing the Traffic Management Plan (i.e., Traffic Control Manager and Traffic Control Supervisor)	

Comments:

Items for follow-up:

Auditor signature: _____ **Date:** _____

Recipient signature: _____ **Date:** _____

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Traffic Management Plan Field Audit Form		
Project Name _____ Project # _____ Location _____ Duration of Work _____ Description of Work _____ Auditor _____ Audit # _____		
FIELD AUDIT – DURING WORK ACTIVITIES Rate the Items and Conditions in the Field Audit form using these three indicators: ✓ acceptable X not acceptable N/A not applicable		
Item	**Conditions**	**Rating**
Traffic Management Plan and Traffic Control Plan	The Traffic Management Plan has been updated as the project progresses (changes to work activities that alter traffic control requirements, changes in work activity scheduling, changes to project initiation and/or completion dates, changes to sub-plans, etc.). The Traffic Management Plan has been modified and the changes have been recorded as the work changes.	
Traffic Control Manager (may be same person as Traffic Control Supervisor)	The Traffic Control Manager has: • exercised full line authority over all onsite Traffic Control Persons • finalized traffic control measures • directed implementation of the Traffic Control Plan • monitored traffic operations to determine the effectiveness of the Traffic Control Plan • overseen Traffic Management Plan modifications necessitated by construction changes and the accommodation of special events • kept the Traffic Management Plan up-to-date	

Item	Conditions	Rating
Traffic Control Supervisor (may be same person as Traffic Control Manager)	The Traffic Control Supervisor is not the same person as the Site Supervisor. The Traffic Control Supervisor is qualified and available at all times (i.e., night-time and during weekends). The Traffic Control Supervisor has ensured that: • the required traffic control devices are in place • the daily traffic control logs are maintained • the work of Traffic Control Persons is coordinated on the work site • Traffic Control Persons are using the required personal protective clothing and equipment • Traffic Control Persons are positioned in safe locations clear of potential environmental hazards (i.e., rock slides and avalanches) • Traffic Control Persons are performing traffic control duties competently and safely	

Item	Conditions	Rating
Traffic Control Persons	Signs associated with the Traffic Control Persons are appropriately displayed, and are covered or removed when not required. The Traffic Control Persons working on the project: • are qualified and carrying up-to-date certification on the work site • are used appropriately • are wearing appropriate personal protective equipment as stipulated in Part 18 of WorkSafeBC's Occupational Health & Safety Regulation (hard hat with retroreflective strip, safety vest, CSA-approved footwear, wrist and ankle bands) • are using standard signals for traffic control as described in Part 18 of WorkSafeBC's Occupational Health and Safety Regulation • are using traffic control paddles that meet the requirements for a C-027 Traffic Control Paddle • are using adequate and effective communication • are using a red signaling wand during night work and in conditions of poor visibility • are standing in the best possible positions unless circumstances or space requirements are restricted • are aware of a clear escape route • are using precise motions to direct traffic • are clearly directing and adequately controlling traffic	

Item	Conditions	Rating
Traffic Control Signs and Devices	Signs are appropriate in size and have diamond-grade retroreflectivity. Signage is adequate, and is spaced appropriately in accordance with the Traffic Management Plan for the project or regulatory speed limits throughout the project. Signage adequately communicates the necessary information to all road users (drivers, pedestrians, and cyclists). Signs are visible, clear of debris, free from obstruction, in good condition, and properly displayed. Sign patterns are displayed consistently in both directions throughout the work zone. Speed drops are appropriate to highway and road speed. Signs and devices are installed in accordance with the Traffic Control Plan. Contradicting signs are covered or removed (e.g., normal 100 km/h sign covered or removed when construction speed is 50 km/h; inapplicable signs covered or removed when work site is inactive). All side roads, intersections, and interchanges have been considered, and are signed accordingly. Delineation is suitable and provides a clear message for directing traffic through the work site. Delineation is properly set up and spaced to provide a suitable buffer. Changeable message boards and speed readers are positioned properly in locations most suitable for project and driver visibility, are in good working condition, and are functioning properly. Night hazards have been addressed. Floodlights have been installed where required and are positioned so that they will not create visibility problems for drivers.	

Item	Conditions	Rating
Traffic Control Layout	Sight distance is adequate as drivers approach the work zone, and at all points within the work zone. Layout is consistent with the Traffic Management Plan and Traffic Control Plan (i.e., detours and road closures are considered, and the proper traffic control devices are being used). Layout is organized and clear. Layout is uniform and consistent to ensure that drivers respond properly. Drivers, pedestrians, and cyclists can move easily through the work zone. Layout is such that it encourages drivers, pedestrians, and cyclists to reduce risk. Standard layouts are used for signage and other traffic control devices to meet traffic control requirements. The work zone has an advance warning area in which drivers can adjust their driving in accordance with the conditions. The transition area is easily identified with the proper traffic control devices (i.e., delineator tapers). The buffer space is adequate and allows drivers to stop at a safe distance. The buffer space is free of equipment, workers, materials, and workers' vehicles, with the exception of a buffer vehicle. The work activity area is closed off to drivers by appropriate traffic control devices (i.e., delineators, barricades, or other channelizing devices). The work activity area provides a safe entrance and exit for work vehicles. The termination area provides an adequate distance for traffic to clear the work zone and return to normal travel lanes.	

Item	Conditions	Rating
Inactive Work Site	The site has been left in a functional condition that allows for the safe movement of all road users, including pedestrians, cyclists, and vehicles. Construction signs that are not being used are covered to prevent driver confusion. Signage and traffic control devices are appropriate. A specific person has been identified as being responsible for 24-hour maintenance and emergency response, and has accepted this responsibility. The traffic control layout is consistent with the Traffic Management Plan regarding inactive work sites.	
Maintenance	The road surface is free from gravel, mud, and other debris. Signage and traffic control devices are properly used, displayed, and set up, and are free from mud, debris, and obstruction. A specific person routinely monitors, inspects, and maintains traffic control signs and devices.	

Comments:

Items for follow-up:

Auditor signature: _____ **Date:** _____

Recipient signature: _____ **Date:** _____

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Appendix E: Lane Closure Request Form

Appendix E contains a sample of the Ministry's Work Notification/Lane Closure Request and Approval Form (H1080).

The online form is available at http://gww.th.gov.bc.ca/forms/getFormFile2.aspx?formId=649.

It can be populated online, and then printed for submission to the Ministry.

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BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

File Number _____
Permit Number _____
Date (yyyy/mm/dd) 20 / /

WORK NOTIFICATION/LANE CLOSURE REQUEST AND APPROVAL

NOTE: This form is to be submitted ten (10) working days prior to start of work or closure.

To _____ (District Manager Transportation) _____ (District Office) Telephone _____ Fax _____

From _____ (Organization) _____ (Contact Person) Telephone _____ Fax _____

☐ I request approval to **work** within the Ministry right-of-way (check as many boxes below as required):

☐ North Bound ☐ South Bound ☐ East Bound ☐ West Bound

☐ I request approval to **close** the Ministry right-of-way (indicate what will be closed by checking ☑ all applicable boxes below):

☐ North Bound ☐ South Bound ☐ East Bound ☐ West Bound

☐ Right Turn Lane ☐ Middle Lane ☐ Left Turn Lane ☐ Work on Shoulder ☐ Cyclists/Pedestrians
☐ Curb/Outside Lane ☐ Centre/Inside Lane ☐ All Lanes ☐ No Lane Closure ☐ Bridges/Intersections

on Highway _____ (Number) between _____ (Landmark) and _____ (Landmark)

from _____ ☐ a.m./ ☐ p.m. to _____ ☐ a.m./ ☐ p.m. on the following dates (yyyy/mm/dd) 20 / /

to 20 / / ; for the purpose of constructing the following works: _____

In the above noted directions, the highway is ☐ one (1) / ☐ two (2) / ☐ three (3) or more lanes wide and
☐ zero (0) / ☐ one (1) / ☐ two (2) lanes of traffic will be maintained in the same directions as the closure.

PLEASE ATTACH THE FOLLOWING:

a) Traffic control diagram figure number _____ as per "Traffic Management Manual for Work on Roadways";
b) A detailed Traffic Management Plan (format available on the Traffic Management for Work on Roadways website).

APPROVAL SECTION

☐ Request denied ☐ Request approved as submitted ☐ Request approved with the following changes:

☐ This approval is granted subject to traffic queues being monitored continuously by the Ministry or Project Representative while lane closures or traffic diversions are under way. Delays are not to exceed _____ minutes over the normal travel time.

☐ This approval is granted subject to traffic queues being monitored continuously by the Ministry or Project Representative while lane closures or traffic diversions are under way. Standards will be set as to what delays are acceptable to the public in the Traffic Management Plan and then work will commence until these standards are not met, at which time the Traffic Management Plan will be adjusted. Following acceptance of the revised Traffic Management Plan, the H1080 form is to be resubmitted for approval.

The Transportation Management Centre BC (TMCBC) 1-866-707-7862 must be contacted as follows:

- **At the installation of the lane closure,**
- **Upon removal of the lane closure.**

This approval must be kept on-hand at the work site.

Traffic Data Program can be accessed from here

_____ (District Manager Transportation (or Designate)) _____ (Date (yyyy/mm/dd))

Once approved, if you have a DriveBC account, you can enter this data directly into DriveBC through the DriveBC Input Tool (DIT). Otherwise, please submit prior to 6:30 a.m. or 24 hours in advance to: Ministry of Transportation and Infrastructure District Office, and cc the Transportation Management Centre BC (TMCBC) at TMCBC@gov.bc.ca. The TMCBC requires 24/7 emergency phone numbers for all projects.

H1080 (2020/02) **Distribution:** Original - District Manager Transportation Copy - Highways Maintenance Contractor

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Appendix F: Tables A to D

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Table C: Risk Evaluation for Emergent and Brief-Duration Work F-5

Table D: Minimum Distances for Mobile Work F-6

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TABLE A – TAPER LENGTHS									
Taper Types (m)		**Regular Posted Speed Limit (km/h)**							
		≤50	**60**	**70**	**80**	**90**	**100**	**110**	**120**
Merge Taper Length	**L_M**	35	55	160	190	210	230	250	280
Lane Shift Taper Length	**L_L**	30	50	80	100	110	120	130	140
Downstream Taper Length	**L_D**	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	**L_S**	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	**L_T**	30	60	160	190	210	230	250	280
Run-In Length on Centreline	**L_R**	40	50	60	60	70	80	90	100

Table A Notes	
Regular Posted Speed Limit	Device spacing and taper lengths should be to the regular posted speed limit.
L_M = Merge Taper Length	Merge length required to close lane on approach to work area. For speeds ≥ 70 km/h, merge length should be at least = $\frac{(\text{lane width of } 3.7 \text{ m}) \times (\text{Posted Speed in km/h})}{1.6}$, rounded to nearest 10 m.
L_L = Lane Shift Taper Length	Used when a lateral shift is needed within the work area. Lane Shift Taper = ½ x L_M, rounded up to nearest 10 m.
L_D = Downstream Taper Length	May be used in work zone termination area to provide a visual cue to drivers that they may return to the original lane or path that was closed.
L_S = TCP, Signal, and Shoulder Taper Length	Shoulder Taper: Used to close shoulders within activity area, or when shoulders might be mistaken for driving lanes. May be increased to 1/3 x L_M on higher-speed highways and freeways where shoulder width is ≥ 2.5 m. Signal and TCP Tapers: Used in advance of a work activity area where traffic is controlled so that the road is used alternately by traffic moving in each direction.

L_T = Minimum Tangent Length between Tapers	Used between successive tapers or at other decision or conflict points to provide time for drivers to become accustomed to the first change and observe traffic control devices for the second change. **L_T** = **L_M**, but for high-speed/high-volume freeways and/or night work, it may be doubled (2 x L_T) to increase time for drivers to become accustomed to the first change.
L_R = Run-In Length on Centreline	May be used on centrelines as minimum tangent length before development of lane departures or lane shifts. Run-in length = 0.8 x speed (in km/h) (US Manual of Uniform Traffic Control Devices).

TABLE B – DEVICE SPACING LENGTHS								
Device Spacing (m)		**Regular Posted Speed Limit (km/h)**						
		≤50	**60**	**70**	**80**	**90**	**100**	**110**	**120**
Construction Sign Spacing	**A**	40	60	80	100	150	150	200	200
Buffer Space	**B**	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	**R**	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	**C**	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	**D**	10	10	30	30	40	40	40	50

Table B Notes	
Regular Posted Speed Limit	Device spacing and taper lengths should be to the regular posted speed limit.
A = Construction Sign Spacing	Recommended minimum spacing for signage. Spacing may be adjusted to accommodate site constraints and/or where high numbers of access points exist. Signs within the work zone should be spaced on the basis of the pre-construction, regulatory speed limit. <u>Maximum Construction Sign Spacing</u>: • Spacing for the sign closest to the work activity area should remain as close as possible to Distance A. • For other construction signs in the advance warning area, spacing may be adjusted up to a maximum distance of 2 x Distance A. • Signs that include a distance measurement (e.g., Construction Ahead Next 2 km) should be placed in accordance with the distance cited on the sign or tab.

B = Buffer Space	The longitudinal distance which provides a margin of safety for both the driver and the workers. It is important that the buffer space be free of equipment, workers, material and vehicles. A buffer vehicle with a crash attenuator may be located within the buffer space if there are space constraints. The buffer space is measured is from the end of the taper to the work activity area unless there is a buffer vehicle, in which case it is measured to the back of the buffer vehicle. Typically used on high-speed roadways but should be considered for all works where space allows. Distance is based on the braking distance on level ground for wet pavement as defined in the Transportation Association of Canada's Geometric Design Guide for Canadian Roads (1999).
R = Roll-Ahead Buffer Distance	The longitudinal distance measured from the front of the buffer vehicle to the work activity area. It provides a margin of safety in case of impact. Distance is based on the Minnesota Manual on Uniform Traffic Control Devices (2011).
C = Channelizing Device Spacing for Tapers	Maximum spacing between channelizing devices for tapers.
D = Channelizing Device Spacing on Curves and Tangents	Maximum spacing between channelizing devices on curves and tangents. Tighter spacing is acceptable especially on curves where device loss can impact directional continuity. Maximum device spacing is calculated as 0.4 x speed (in km/h), rounded to the nearest 10 m.

TABLE C – RISK EVALUATION FOR EMERGENT OR BRIEF-DURATION WORK

Risk Evaluation Category	Risk Criteria	Criteria Met?
1. Work Duration	Can the work be completed in 5 minutes or less?	Yes / No
2. Sight Distance Distance from parked location to furthest point that can be seen on the road.	For the posted speed limit, is the minimum sight distance met?	Yes / No

Speed Limit (km/h)	50 - 70	80 - 90	100 - 110	120
Minimum Sight Distance (m)	100	170	250	300

3. Traffic Volume	Is the traffic volume in lanes that will be entered by workers estimated to be less than 5 vehicles per lane per minute?	Yes / No
4. Environmental Conditions	Is visibility unrestricted (no fog, blowing snow, etc.) **and** are road conditions not slippery?	Yes / No

Table C Notes

A Risk Evaluation has three possible outcomes:

1. **Answers to all risk criteria questions are Yes:** Traffic control devices may be installed in accordance with the appropriate Emergent Work traffic control layout.

2. **Answers to one or two risk criteria questions are No:** Traffic control devices may be installed in accordance with the appropriate Brief-Duration Work traffic control layout.

3. **Answers to three or more risk criteria questions are No:** Additional traffic control measures are required beyond those described and illustrated for Emergent and Brief-Duration Work. The standard layout(s) for the appropriate short-duration, long-duration, or mobile work should be applied instead, which may necessitate calling in additional resources.

TABLE D – MINIMUM DISTANCES FOR MOBILE WORK						
	Regulatory Speed Limit (km/h)					
	≤50	60	70 - 80	90	100	≥ 110
Minimum Distance Moved (m) Every 30 Minutes	100	130	170	220	260	300

Table D Notes

To be considered mobile work, the operation must move at least the distance shown in Table D for the posted speed limit every 30 minutes or less.

If the work does not regularly move the specified distance, it should be treated as a stationary operation, and the appropriate layout should be used.